

A light blue background featuring a network diagram. It consists of several circular nodes, each containing a stylized human silhouette. These nodes are interconnected by thin, light blue lines, forming a complex web that symbolizes data communication and networks.

Data Communication and Networks

Contents

- Introduction to Data Communication
- Data Communication Vs. Computer Network
- Network Components
- Data Representation
- Data Flow
- Network Physical Structure
- Network Physical Topology



Introduction



- Data communications and networking have changed the way we do business and the way we live
- Information on fingertips, Google, ChatGPT
- Internet applications like YouTube, WhatsApp, Facebook, Instagram etc
- Email, File Sharing, ERP

Introduction



- When we communicate, we are sharing information.
- Sharing can be local (**face-to-face**) or remote (**telecommunication**).
- The term Telecommunication, which includes *telephony*, *telegraphy*, and *television*, means communication at a distance.
- The word Data refers to information presented in whatever form is agreed upon by the parties creating and using the data.
- Data communications (DC) are the exchange of data between two devices via some form of transmission medium such as a wired cable or a wireless medium.
- Communicating devices must be part of a communication system made up of a combination of *hardware* (physical equipment) and *software* (programs)

Computer Network

- A **computer network** is a collection of **interconnected computers and other devices** that can communicate with each other and share resources.
- A network may contain computers, servers, switches, routers, access points, printers, IoT devices, etc.

Data Communication and Computer Network

- **Data Communication** = "How data is exchanged."
- **Computer Network** = "The connected system through which data can be exchanged."
- Thus, **data communication** is a fundamental function of **computer networks**. Without data communication, a computer network would simply be a collection of connected devices with no meaningful exchange of information.

Difference: Data Communication and Computer Network

Aspect	Data Communication	Computer Network
Meaning	Exchange of data between devices	Interconnection of devices
Main focus	Transmission of data	Connectivity and resource sharing
Components	Sender, receiver, message, medium, protocol	Computers, switches, routers, cables, protocols, etc.
Scope	Can occur between two devices	Usually involves multiple interconnected devices
Example	Sending a file from PC A to PC B	University LAN connecting hundreds of computers

Introduction Contd..

- The effectiveness of a data communications system depends on four fundamental characteristics:

- **Delivery**: Data delivery to the correct destination.



- **Accuracy**: System must deliver the data accurately.



- **Timeliness**: System must deliver data in a timely manner.



- **Jitter**: Refers to the variation in the packet arrival time (uneven delay in delivery).

DC/Network Components

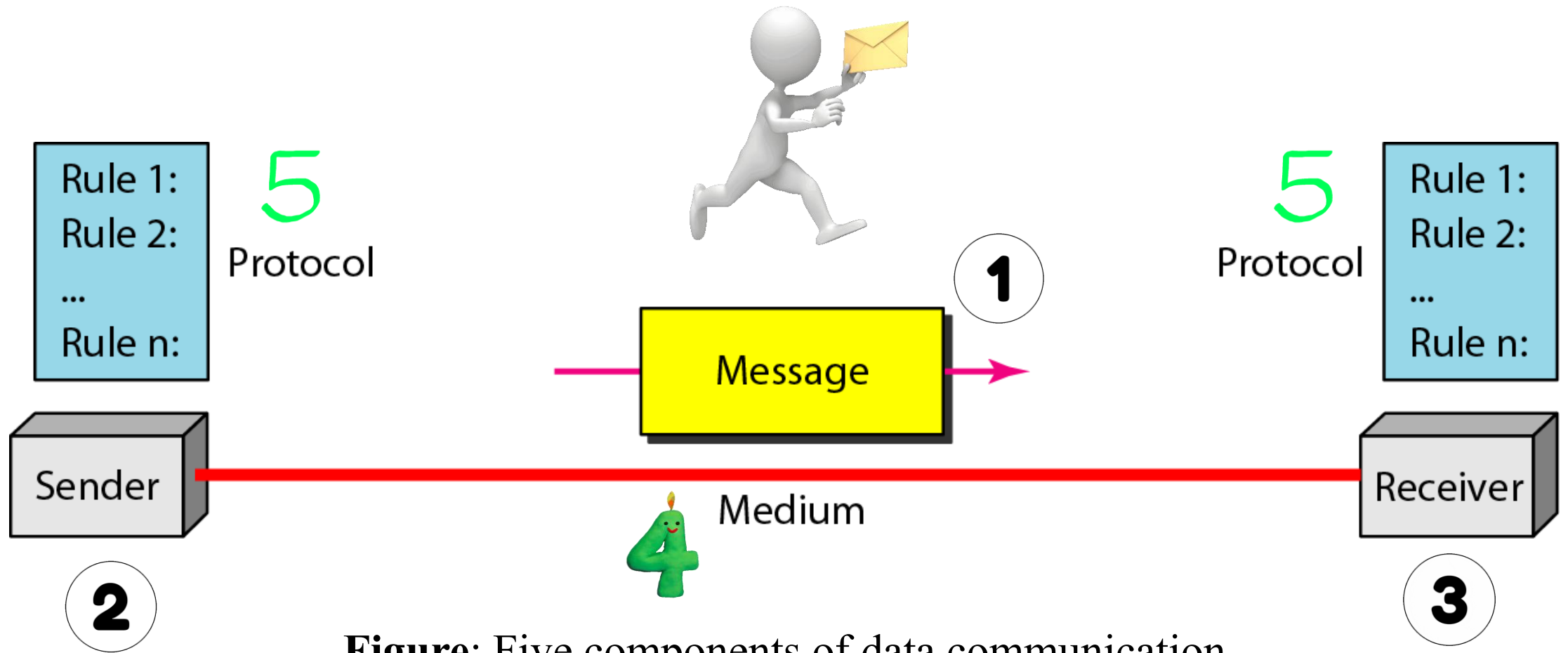


Figure: Five components of data communication

Data Representation

- Information today comes in different forms such as text, numbers, images, audio, and video.
- Text:
 - Sequence of bits (**0s or 1s**).
 - Different sets of **bit patterns** have been designed to represent text symbols.
 - Each set is called a **code**, and the process of representing symbols is called **coding**. Code eg Unicode and ASCII
- Numbers:
 - Numbers are also represented by **bit patterns**.



16 8 4 2 1
1 0 1 1 1

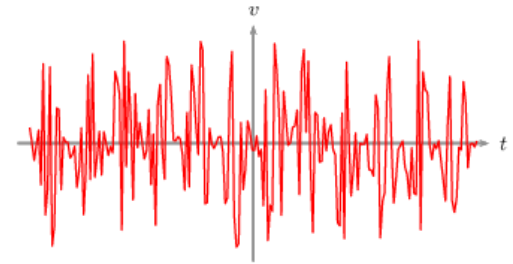
Data Representation

- Images:

- Images are also represented by bit patterns.
- Composed of a matrix of pixels. Each pixel represented by bit pattern
- Monochrome image pixel represents black or white color. 1-bit is sufficient for each pixel/
- 2-bit pattern (00-Black and 11-White) for **BW images** and RGB/CMY/HSV for **coloured images**.

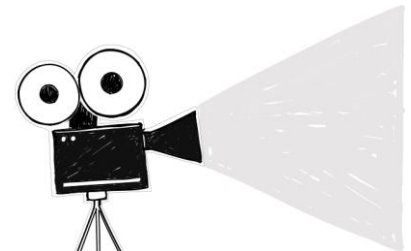
- Audio:

- Audio refers to the recording or broadcasting of sound or music.
- It is **continuous**, not discrete.



- Video:

- Video refers to the **recording or broadcasting** of a picture or movie.
- Video can either be produced as a **continuous entity**, or it can be a combination of images.

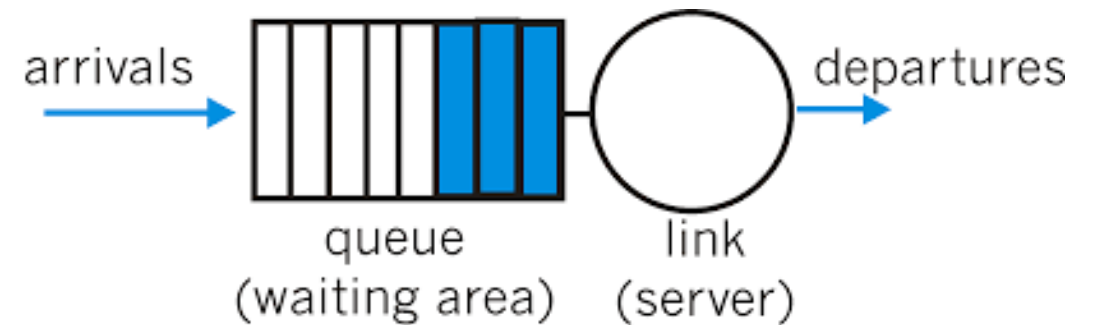
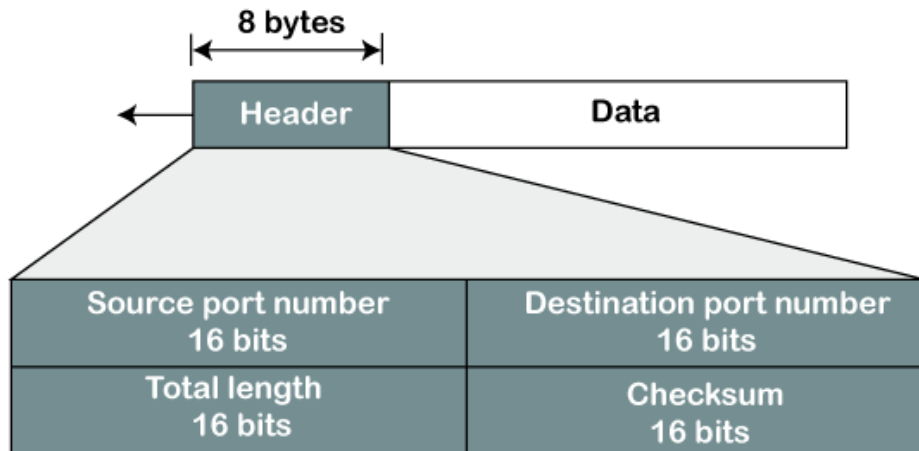


Network Communication Protocols

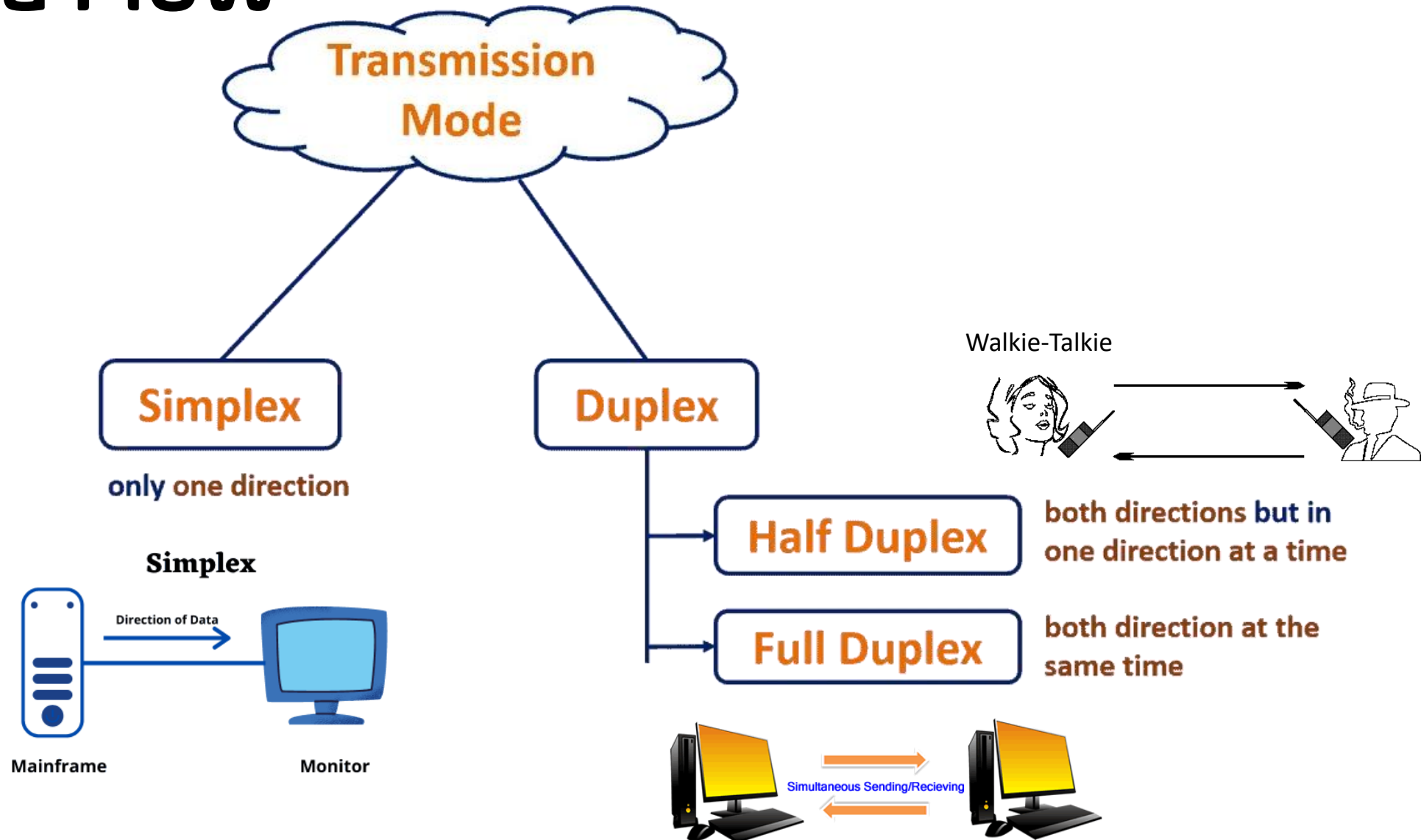
- In computer networks, communication occurs between entities in different systems.
- However, two entities **cannot simply send** bit streams to each other and expect to be understood.
- For communication to occur, the entities must **agree on a protocol**.
- A protocol is a **set of rules** that govern data communications.
- A protocol defines **what** is communicated, **how** it is communicated, and **when** it is communicated.
- The key elements of a protocol are **syntax, semantics, and timing**.

Network Communication Protocols

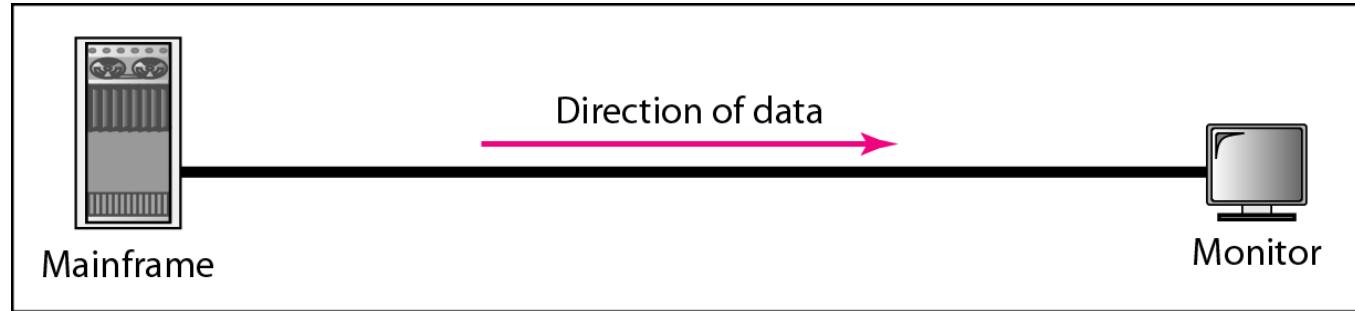
- **Syntax:** Structure or format of the data.
- **Semantics:** Refers to the meaning of each section of bits.
- **Timing:** When data should be sent and how fast they can be sent. For example, if a sender produces data at 100 Mbps but the receiver can process data at only 1 Mbps, the transmission will overload the receiver and some data will be lost.



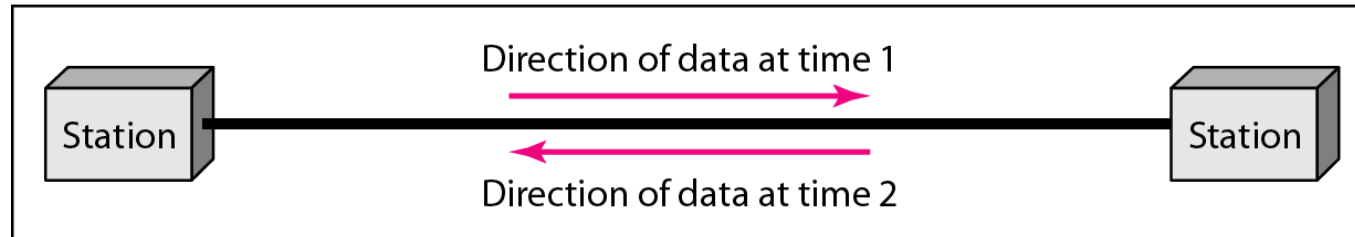
Data Flow



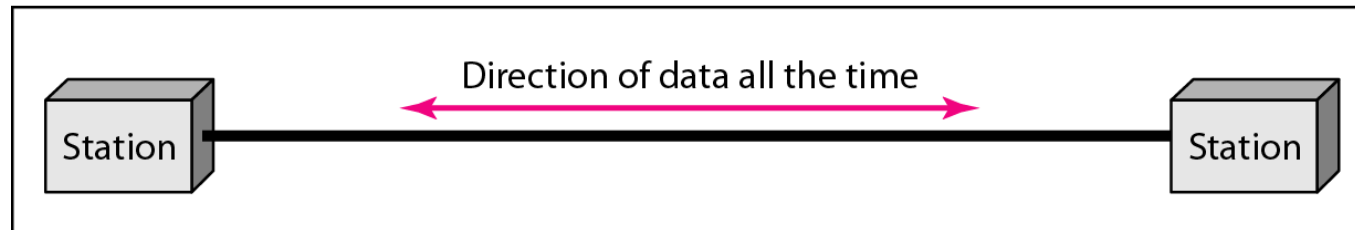
Data flow (simplex, half-duplex, and full-duplex)



a. Simplex



b. Half-duplex



c. Full-duplex

Simplex Communication

- In **simplex communication**, data travels **only from sender to receiver**. The receiver cannot send data back through the same communication channel.
- **Applications:**
 - **Television broadcasting** – TV station → TV
 - **Radio broadcasting** – Radio station → Radio receiver
 - **Keyboard → Computer** – traditionally considered a simplex example
 - **Computer → Printer** – traditionally, data is sent from computer to printer
 - **Digital signage** – central system → display
 - **Satellite broadcasting** – satellite → receiving station
 - **Public announcement systems** – announcer → audience

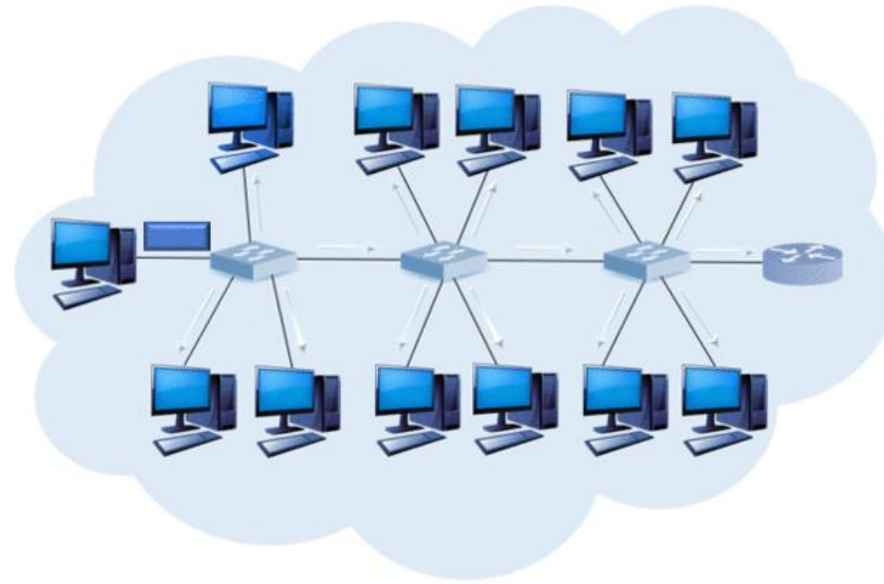
Half-Duplex Communication

- In **half-duplex communication**, both devices can **send and receive**, but **not at the same time**.
- **Applications:**
 - **Walkie-talkies** – one person speaks while the other listens
 - **Police radio communication**
 - **Firefighter communication systems**
 - **Aircraft radio communication** – traditionally uses push-to-talk communication
 - **Marine radio communication**
 - **Some two-way radio systems**
 - **Older Ethernet networks using hubs** – devices share the medium and cannot transmit simultaneously without collisions

Full-Duplex Communication

- In **full-duplex communication**, both devices can **send and receive data simultaneously**.
- **Applications:**
 - Telephone calls
 - Mobile phone conversations
 - Video conferencing
 - VoIP communication
 - Modern switched Ethernet
 - Serial communication using separate transmit and receive lines
 - Remote desktop applications
 - Two-way video communication
 - Some vehicle communication systems

Network



- A network is a **set of devices** (often referred to as nodes) connected by communication links.
- A node can be a computer, printer, or any other device **capable of sending and/or receiving** data generated by other nodes on the network.
- Device can be a host or a connecting device, connected using **wired or wireless** medium.
- A network must be able to meet a certain number of criteria (*performance, reliability, and security*).

Network Criteria

- **Performance**

- Transit time, response time, number of users, type of transmission medium, capabilities of the connected hardware, efficiency of software, throughput, and delay.

- **Reliability**

- Frequency of failure, time taken for a link to recover from a failure, network's robustness in catastrophe.

- **Security**

- Protecting data from unauthorized access, protecting data from damage and development, implementing policies and procedures for recovery from breaches and data losses.

Network: Physical Structures

- A network is two or more devices connected through links, so it must have some structure.
- Two types of connections

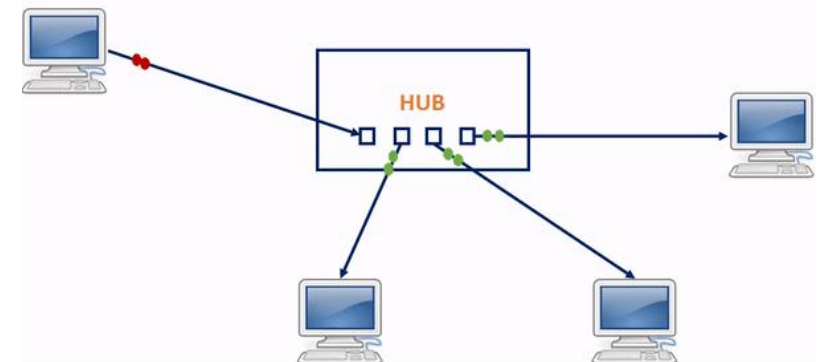
- **Point-to-point connection**

- Dedicated link between two devices; entire capacity is reserved for the two devices
- Examples:
 - A leased line between two offices.
 - A direct fiber optic connection between two routers.



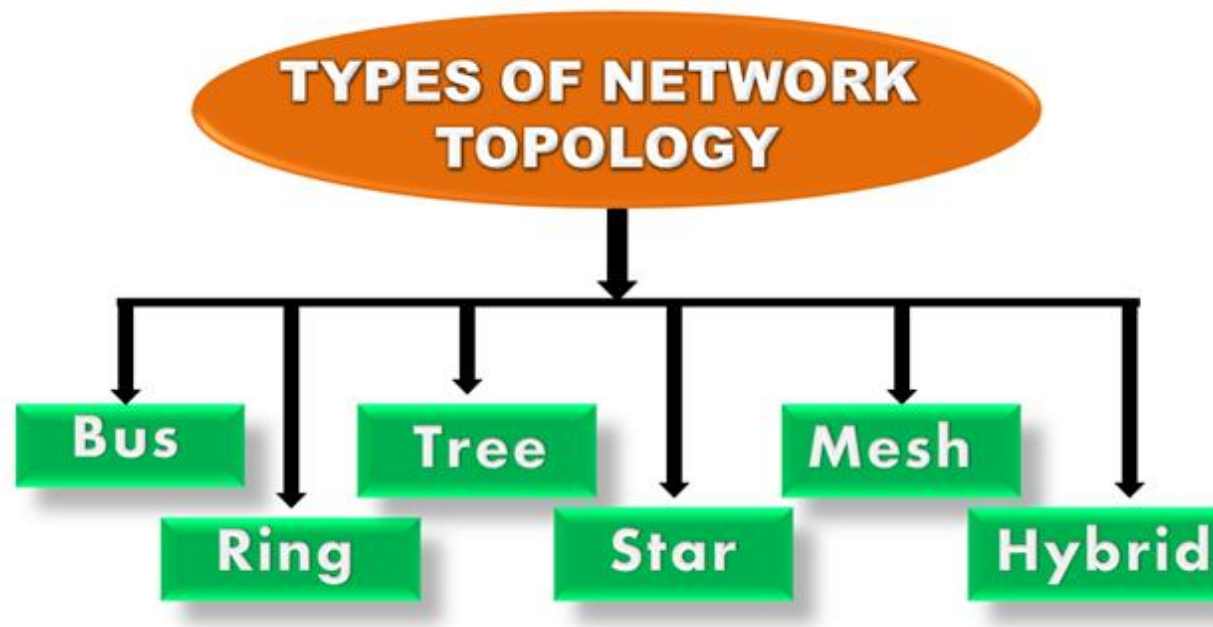
- **Multipoint connection**

- More than two specific devices share same link
- Channel capacity is shared
- Examples:
 - A wireless access point serving multiple clients.
 - Satellite uplink transmitting to multiple ground stations.



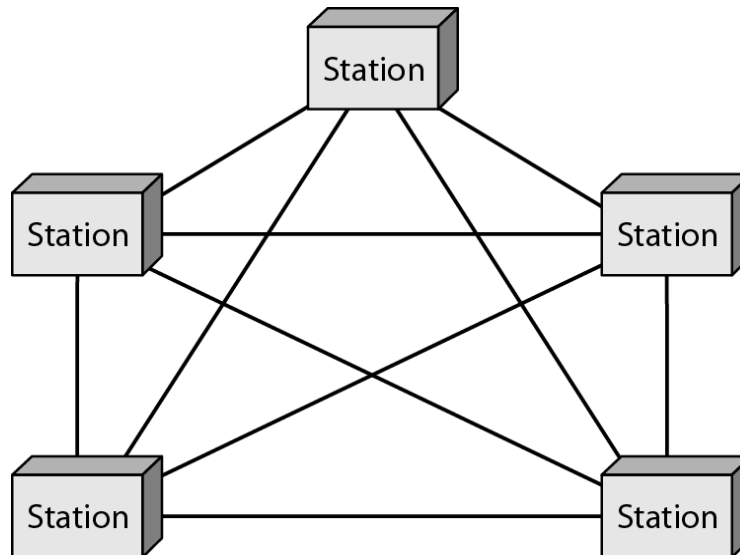
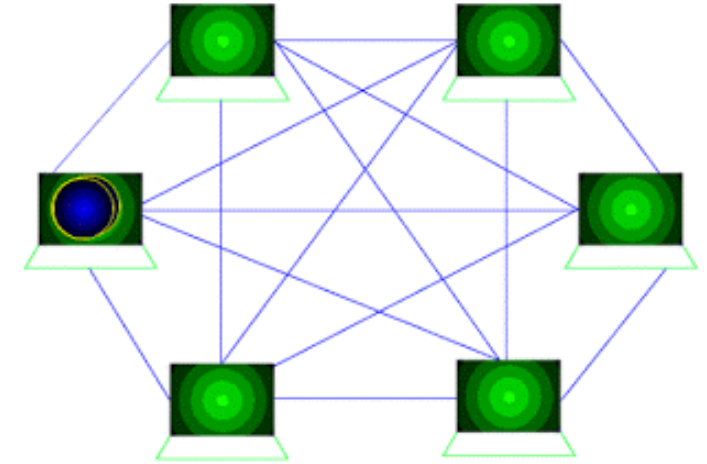
Network: Physical Topologies

- Way the network is laid physically.
- Geometric representation of the relationship of all links and linking devices (nodes) to one another.



Mesh Topology

- **Dedicated** link to all nodes.
- Number of physical links (if n nodes):
 - $n(n-1)$ – if simplex mode (two links for up and down)
 - $n(n-1)/2$ – if full-duplex mode
- *Example:* Each telephone regional office is connected to every other regional office



Advantages of Mesh topology

- High reliability**

If one link fails, data can use another available path.

- Fault tolerance**

Failure of a single device or connection usually does not bring down the entire network.

- Dedicated connections**

In a full mesh, each pair of devices has a dedicated communication link, reducing congestion.

- Better security and privacy**

Dedicated links can provide greater isolation between communicating devices.

- Easy fault identification**

Since devices have separate links, it is relatively easier to identify which connection has failed.

- High performance**

Multiple paths allow data to travel through alternative routes, which can improve network availability and performance.

- Suitable for critical applications**

Mesh topology is useful where network availability is very important, such as **WAN backbones, telecommunications, and critical infrastructure.**

Disadvantages of Mesh topology

- **High installation cost**

A full mesh requires a large number of cables and network interfaces.

- **Complex installation**

Designing and connecting many links becomes complicated as the number of devices increases.

- **Difficult maintenance**

Troubleshooting and maintaining a large number of connections can be challenging.

- **Large amount of cabling**

Full mesh requires many physical connections, resulting in more cables and infrastructure.

- **Requires more ports**

Each device in a full mesh needs a separate interface for every other connected device.

- **Not easily scalable**

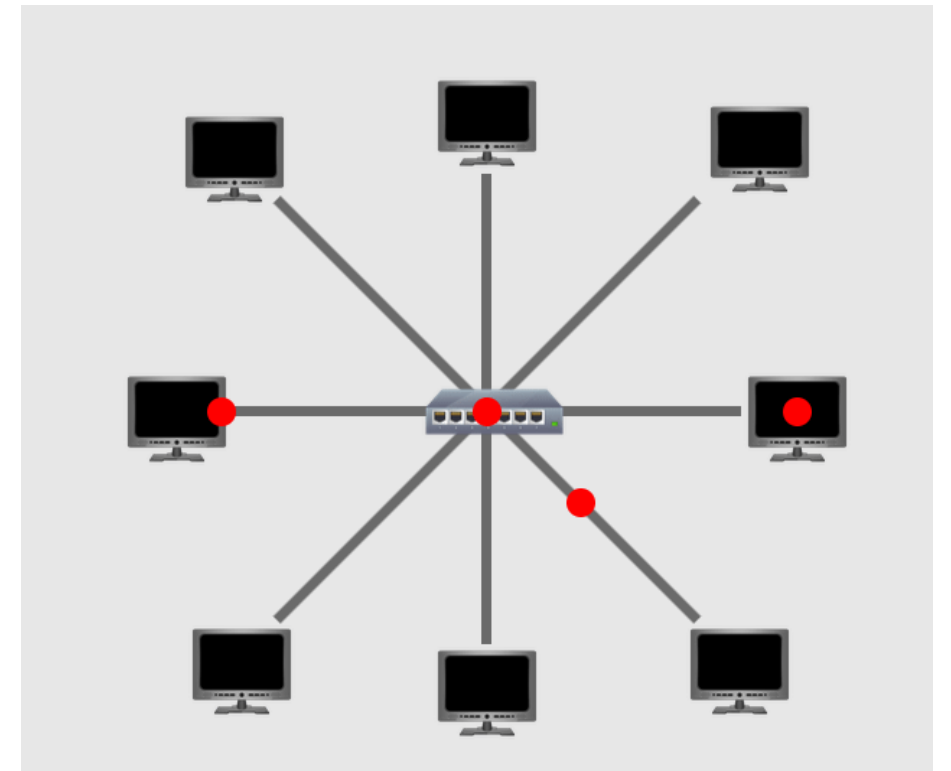
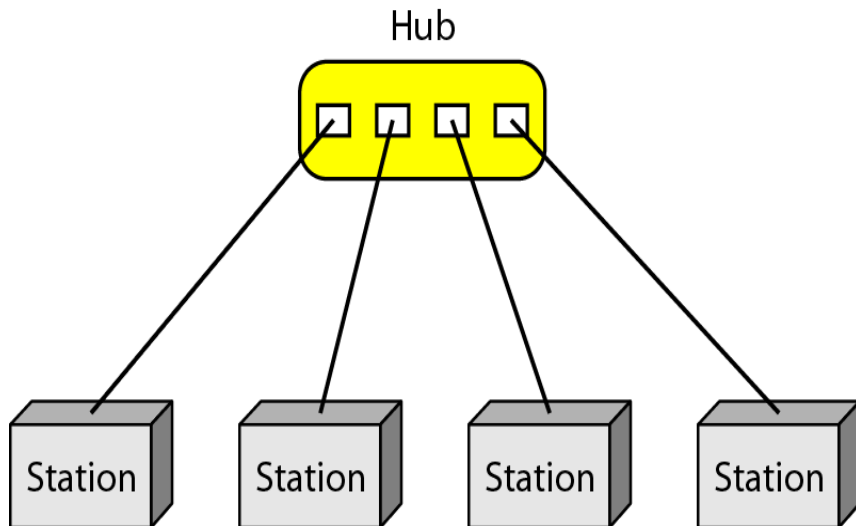
Adding a new device to a full mesh requires additional connections to existing devices.

- **Higher management complexity**

Managing routing, links, and network configurations becomes increasingly difficult as the network grows.

Star Topology

- Each device has **dedicated** link with the central controller (hub).
- Does not allow direct traffic between devices
 - Controller relays the traffic.
- Its dependency on one single point.
- Example: LANs.



Advantages of Star topology

- **Easy to install**

Connecting each device directly to a central switch makes network setup relatively simple.

- **Easy to manage**

The central switch makes it easier to monitor and manage connected devices.

- **Easy fault detection**

If one cable or computer fails, it is usually easy to identify the affected connection.

- **Failure of one device does not affect the others**

If one computer or its connecting cable fails, the remaining devices can continue communicating.

- **Easy to expand**

New computers or devices can be added by connecting them to an available port on the central switch.

- **Better performance**

When a **switch** is used, devices generally have dedicated connections to the switch, reducing unnecessary traffic and collisions.

- **Good network security**

The central switch can be configured with security features such as port security and access controls.

- **Easy troubleshooting**

Since each device has a separate connection to the central device, troubleshooting is generally easier than in bus or ring topologies.

Disadvantages of Star topology

- Central device is a single point of failure**

If the central switch or hub fails, **communication among all connected devices is disrupted.**

- Higher cost**

It requires a separate cable connection from every device to the central device.

The cost of cables and the central switch can therefore be higher than some other topologies.

- More cabling**

Each computer must have its own cable connecting it to the central device.

This can require a considerable amount of cabling in a large network.

- Dependence on the central device**

Network performance and connectivity depend heavily on the central switch or hub.

- Limited by the number of ports**

A switch has a fixed number of ports. If all ports are occupied, additional devices require another switch or expansion equipment.

- Central device can become a bottleneck**

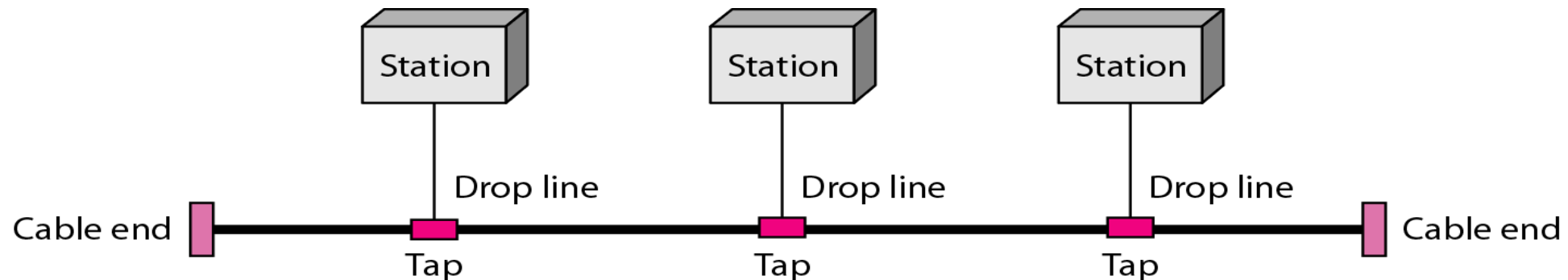
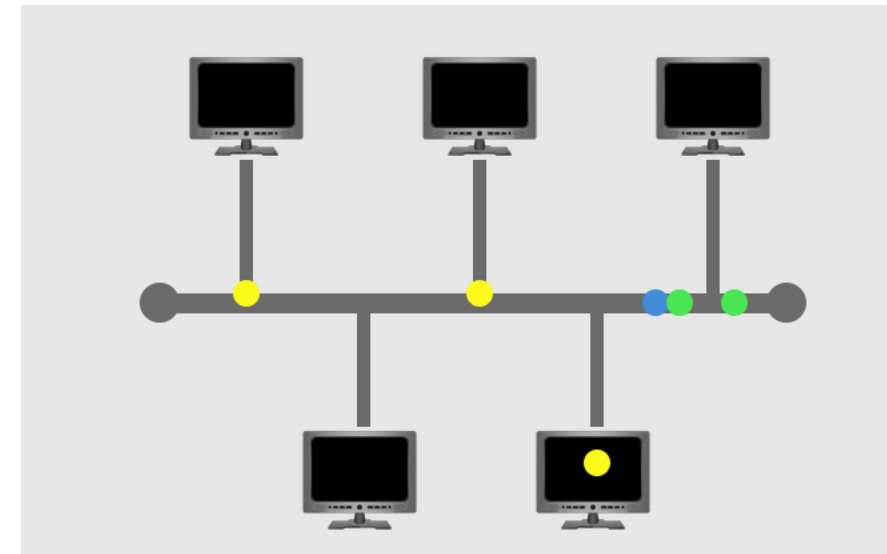
If the central switch does not have sufficient capacity, heavy traffic from many devices can affect network performance.

- Installation can be difficult in large networks**

Running individual cables from every device to a central location can be challenging, particularly in large buildings.

Bus Topology

- **Multipoint** topology.
- One long cable acts as a **backbone** to link all devices.
- Nodes connected by **drop lines** and **taps**
 - *Drop line* is connected between main line and node.
 - *Tap* is a connector that splices or punctures into main cable to create contact with the metallic core.
- Difficult to reconnect and manage faulty installation.



Advantages of Bus topology

- Easy to install**

The network uses a single main cable, making the initial setup relatively simple.

- Low cost**

It requires less cable than star or mesh topology, especially for a small network.

Fewer networking devices may be required.

- Less cabling**

All devices share the same backbone cable, so individual cables from every computer to a central switch are not required.

- Easy to extend for small networks**

A new device can be connected to the existing backbone, provided the cable and network capacity support the addition.

- Does not require a central device**

Unlike star topology, a bus network does not require a central hub or switch for basic communication.

- Suitable for small networks**

It can be useful where there are a small number of devices and network traffic is relatively low.

- Simple structure**

The topology has a straightforward physical arrangement, making it easy to understand

Disadvantages of Bus topology

- Backbone cable failure affects the entire network**

If the main backbone cable breaks, communication across the network can be disrupted. This is one of the most serious disadvantages of bus topology.

- Difficult to troubleshoot**

Finding the exact location of a cable fault can be difficult because all devices share the same backbone.

- Performance decreases with more devices**

As more devices are connected, more devices compete for the shared communication medium. Network performance can therefore decrease as network traffic increases.

- Data collisions can occur**

When multiple devices transmit data at the same time, their signals can collide. Traditional bus Ethernet used **CSMA/CD** to detect and handle such collisions.

- Limited cable length**

The backbone cannot be extended indefinitely because signal quality decreases over long distances.

- Limited number of devices**

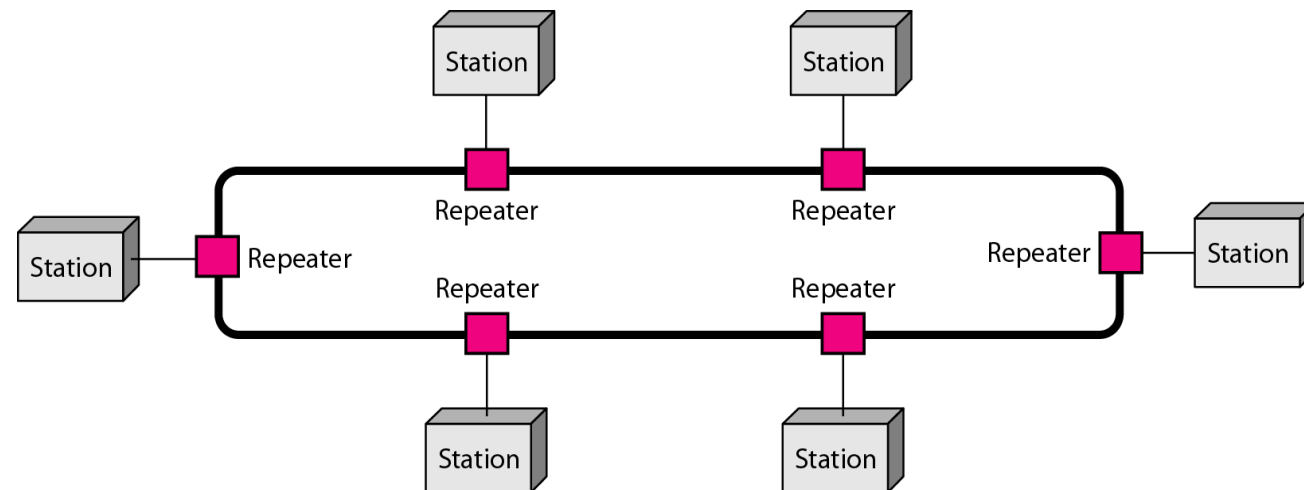
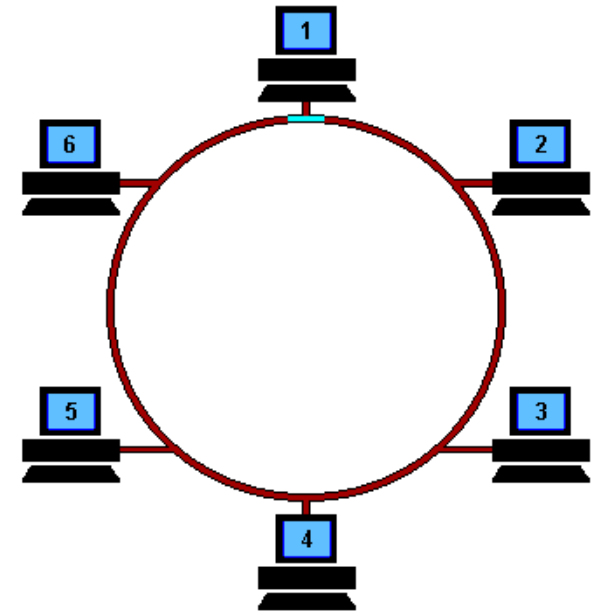
Only a certain number of devices can be effectively connected to the backbone. Adding too many devices increases traffic and can reduce performance.

- Security is relatively weak**

Since devices share the same communication medium, transmitted data may potentially be accessible to other devices on the bus.

Ring Topology

- Each device has a dedicated point-to-point connection with only two devices on either side of it
- When a device receives a signal intended for another device, its repeater regenerates bits and passes them along.



Advantages of Ring topology

- **Less data collision**

- In traditional ring networks, devices may use **token passing**, which controls when a device can transmit.
- This significantly reduces the possibility of data collisions.

- **Efficient data transmission**

- Data travels through a predefined path from one device to another.
- This can provide predictable network performance.

- **Equal access to the network**

- With token-passing mechanisms, each device gets an opportunity to transmit data.
- No single device can continuously dominate the network.

- **Good performance under heavy traffic**

- Ring networks can perform better than shared-medium networks such as traditional bus networks when traffic increases.

- **No central device required**

- A basic ring topology does not require a central hub or switch.
- Each device communicates through its neighboring devices.

- **Easy to identify the direction of data flow**

- Data generally follows a defined direction around the ring, making the transmission path predictable.

- **Suitable for long-distance communication**

- In some ring-based network technologies, signals can be regenerated at intermediate devices, allowing communication over relatively long distances.

- **Relatively equal network load**

- Since devices participate in forwarding data, the communication workload can be distributed across the ring.

Disadvantages of Ring Topology

- Failure of one device or link can affect the network**

- In a traditional single-ring topology, if one device or connection fails, the ring may be broken and communication can be disrupted.

- Difficult to troubleshoot**

- Identifying the faulty device or cable can be more difficult because devices are connected in a circular arrangement.

- Difficult to add or remove devices**

- Adding or removing a device can interrupt network operation because the ring connection needs to be maintained.

- Network expansion is difficult**

- Increasing the number of devices can make the network more complex and may affect performance.

- Data travels through intermediate devices**

- A packet may need to pass through several devices before reaching its destination, increasing the delay.

- Failure can cause significant disruption**

- In a **single ring**, one break can potentially affect communication for many or all devices.

- More complex maintenance**

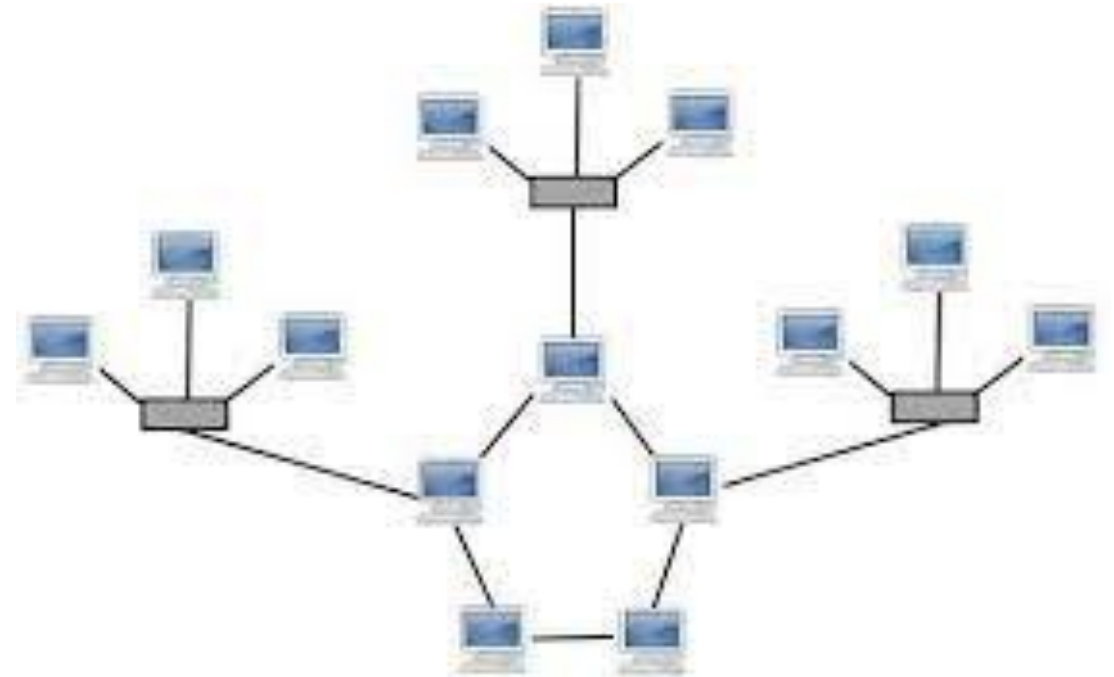
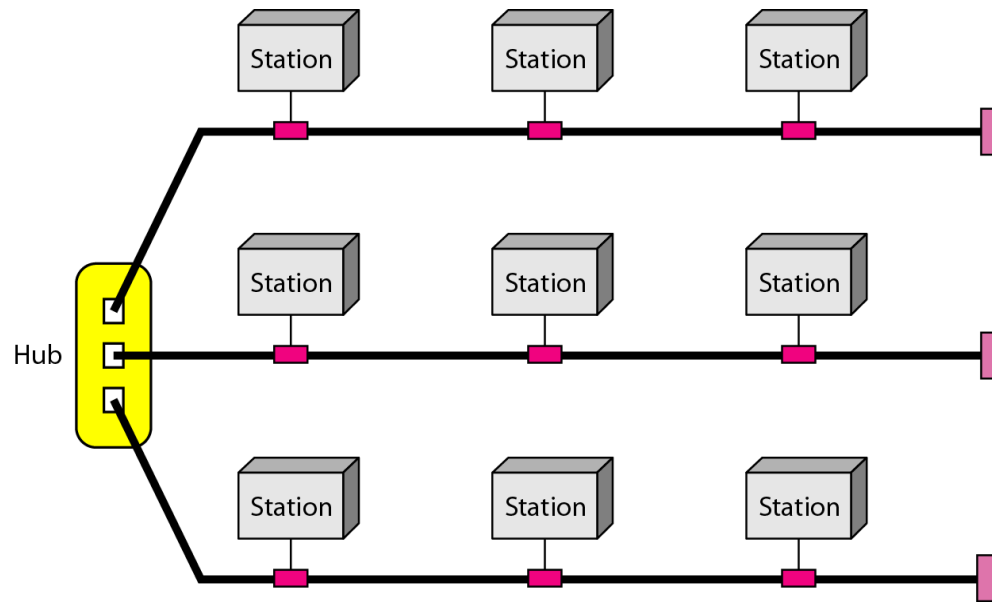
- Maintaining the ring and ensuring that every connection works correctly can be more complicated than maintaining a star network.

- Limited flexibility**

- The physical arrangement of devices must follow the ring structure, making network modifications less flexible.

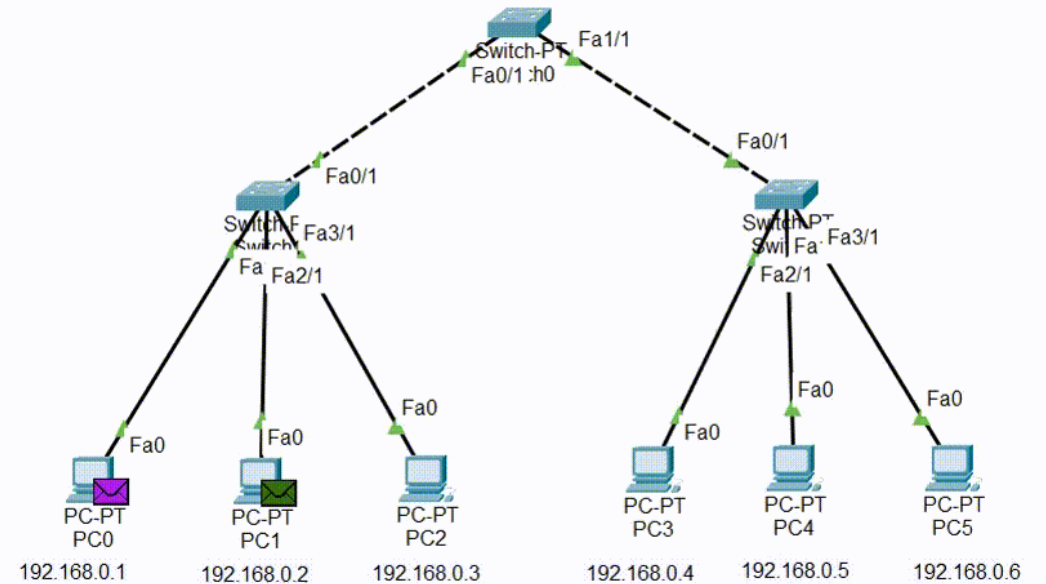
Hybrid Topology

- A network can be hybrid. For example, we can have a main star topology with each branch connecting several stations in a bus topology.



Tree Topology

- A tree topology is a sort of structure in which each node is related to the others in a **hierarchy**.
- Sometimes it is also called hierarchical topology as in this topology, all elements are arranged like the branches of a tree.
- Tree topologies are commonly used to arrange data in **databases and workstations** in corporate networks.
- Tree topology, can be viewed as a collection of star networks arranged in a hierarchy.



Advantages of Tree topology

- Highly scalable**

- New devices, branches, or network segments can be added without redesigning the entire network.
- It is suitable for **large networks**.

- Easy network management**

- Devices can be organized into different branches or hierarchical levels.
- Each branch can be managed independently.

- Easy fault identification**

- The hierarchical structure helps administrators locate the affected branch or segment when a problem occurs.

- Easy expansion**

- Additional switches or network segments can be connected to existing branches.
- This makes tree topology more flexible than a simple bus or ring topology.

- Supports hierarchical organization**

- Devices can be grouped according to **departments, buildings, floors, or functions**.
- For example, a university can have separate branches for CSE, MCA, administration, and laboratories.

- Suitable for large networks**

- Because devices are divided into branches, tree topology can accommodate a large number of devices.

- Better network control**

- Network administrators can control different branches separately and implement different policies for different segments.

- Failure isolation**

- A failure in one lower-level branch may affect only that branch rather than the entire network, provided the main/root connections remain operational.

Disadvantages of Tree topology

- **Backbone failure can affect the entire network**

- If the main backbone or root-level device fails, communication between multiple branches may be disrupted.

- **Higher cost**

- Tree topology requires multiple switches, routers, cables, and other networking equipment.

- Therefore, it can be more expensive than simple bus or ring topologies.

- **Complex installation**

- Designing and installing a hierarchical network with multiple levels and branches can be complicated.

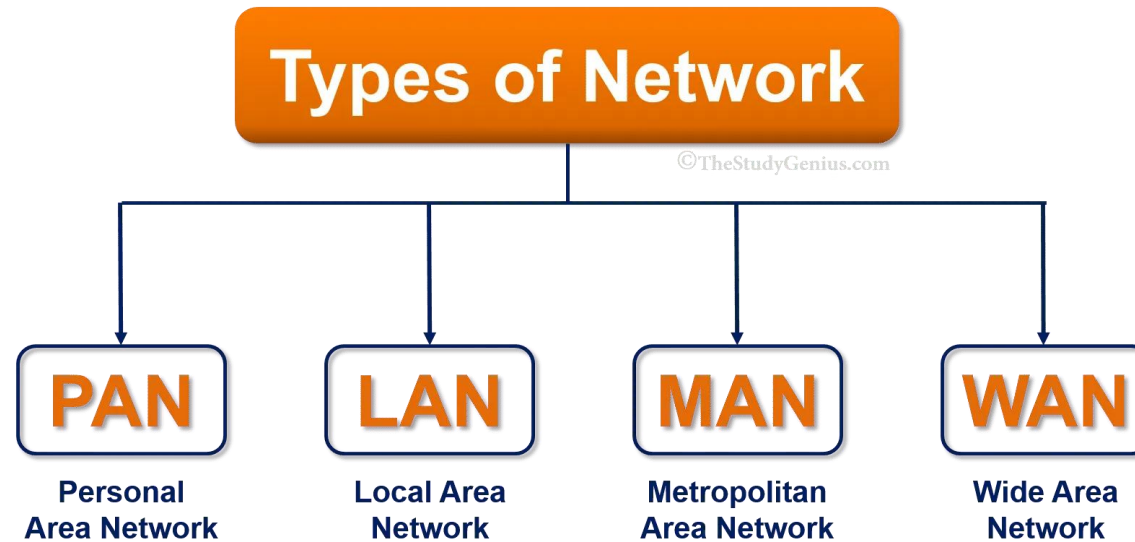
- **Configuration is more complex**

- Network administrators need to configure and manage multiple switches, routers, subnets, VLANs, and links.

- **Performance can depend on higher-level devices**

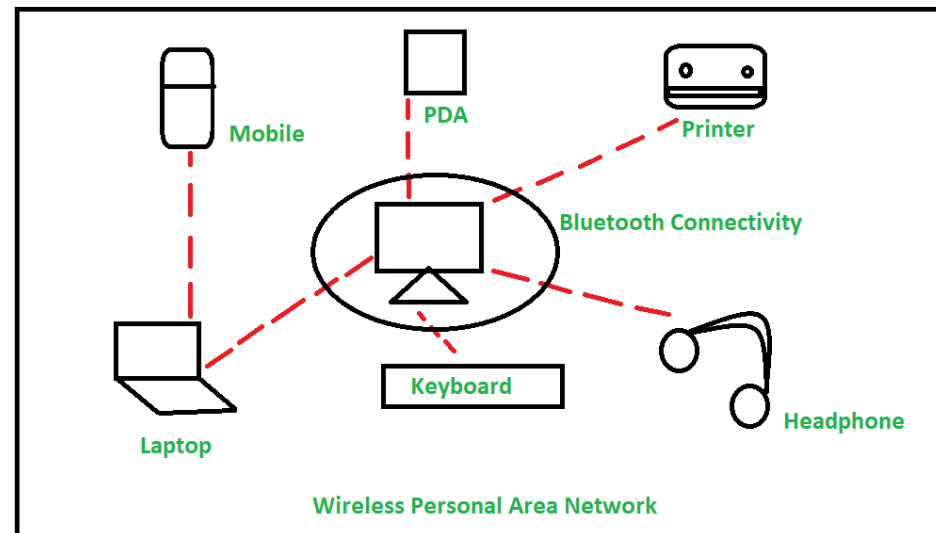
- If a higher-level switch or connection becomes overloaded, the performance of the branches connected through it may be affected.

Categories of Networks



Personal Area Network

- It is a network which connects devices within small range typically on the order of 10 to 100 meters.
- It involves a computer, phone, tablet, printer, PDA (Personal Digital Assistant) and other and other entertainment devices like speakers, video game consoles, etc.
- PAN is relatively flexible and provides high efficiency for short network ranges.

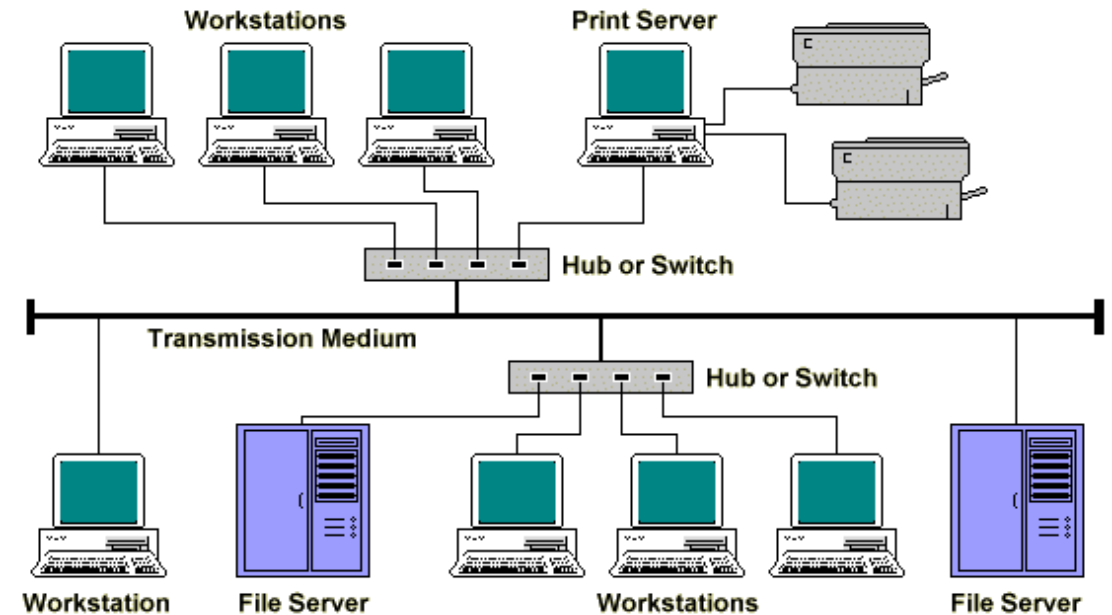


Personal Area Network

- Networks meant for one person
- E.g. Connecting computer with Printer
- Area covered upto one room

Local Area Network

- Covers a **small region** of space, typically a single building.
- The simplest form of LAN is to connect two computers together.
- LAN is operated within a **limited physical area** such as at home, school, a single building or several buildings.
- A network which consists of less than few hundred interconnected devices across several buildings.
- LAN is very **high-speed** network.



Local Area Network

- Privately owned networks within single building or campus
- Area covered upto few kilometers
- Generally used by business units, educational institutions etc
- Types of LAN characterized by
 - Size
 - Knowing size limit in advanced, network design becomes easy
 - Transmission technology
 - Types of transmission medium (which types of cables or other wireless technologies)
 - Speed of traditional LANs : 10 Mbps to 100 Mbps
 - In today's era LANs speed: upto 10 Gbps
 - Topology

Local Area Network

- Types of LAN characterized by
 - Size
 - Transmission technology
 - Topology
 - Various topologies are possible for Broadcast LANs
 - E.g. Bus topology, Ring topology
 - At given point of time only machine can transmit data and others have to refrain from transmission
 - In case of collision in transmission, each machine needs waits for random amount of time and then try again for transmission
 - E.g. IEEE 802.3 known as Ethernet which operates at 10 Mbps to 10 Gbps
 - E.g. IEEE 802.5 (the IBM token ring) which operates at 4 to 16 Mbps
 - Based on channel allocation methods, broadcast LANs can be static or dynamic
 - Static Broadcast LANs allocates channel to each machine in round robin fashion but this method may waste channel capacity in case of nothing to transmit from machine who has access to channel for that time duration
 - Dynamic Broadcast LANs can have centralized or decentralized control mechanism for allocating LANs
 - In case of centralized mechanism algorithm decides which machine will have access to channel next for transmission
 - In case of decentralized channel allocation method each machine decides its own whether to transmit or not

Metropolitan Area Network

- The Metropolitan Area Network (MAN) is a network type that covers the network connection of an entire city or connection of a small area.
- Collection of LANs with the same geographical area, for instance a city.
- The area covered by the network is connected using a wired network, like data cables.
- MAN often acts as a high speed network (although not as fast as LAN) to allow sharing of regional resources.

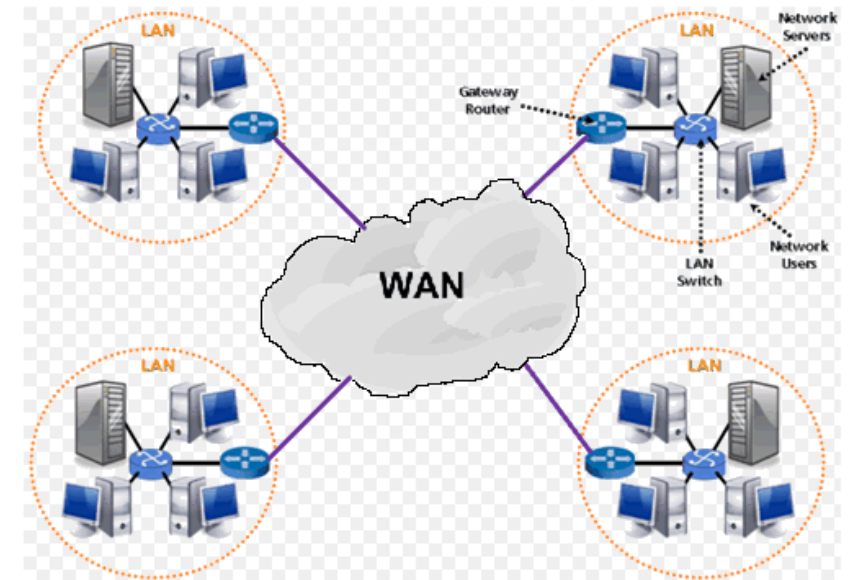


Metropolitan Area Network

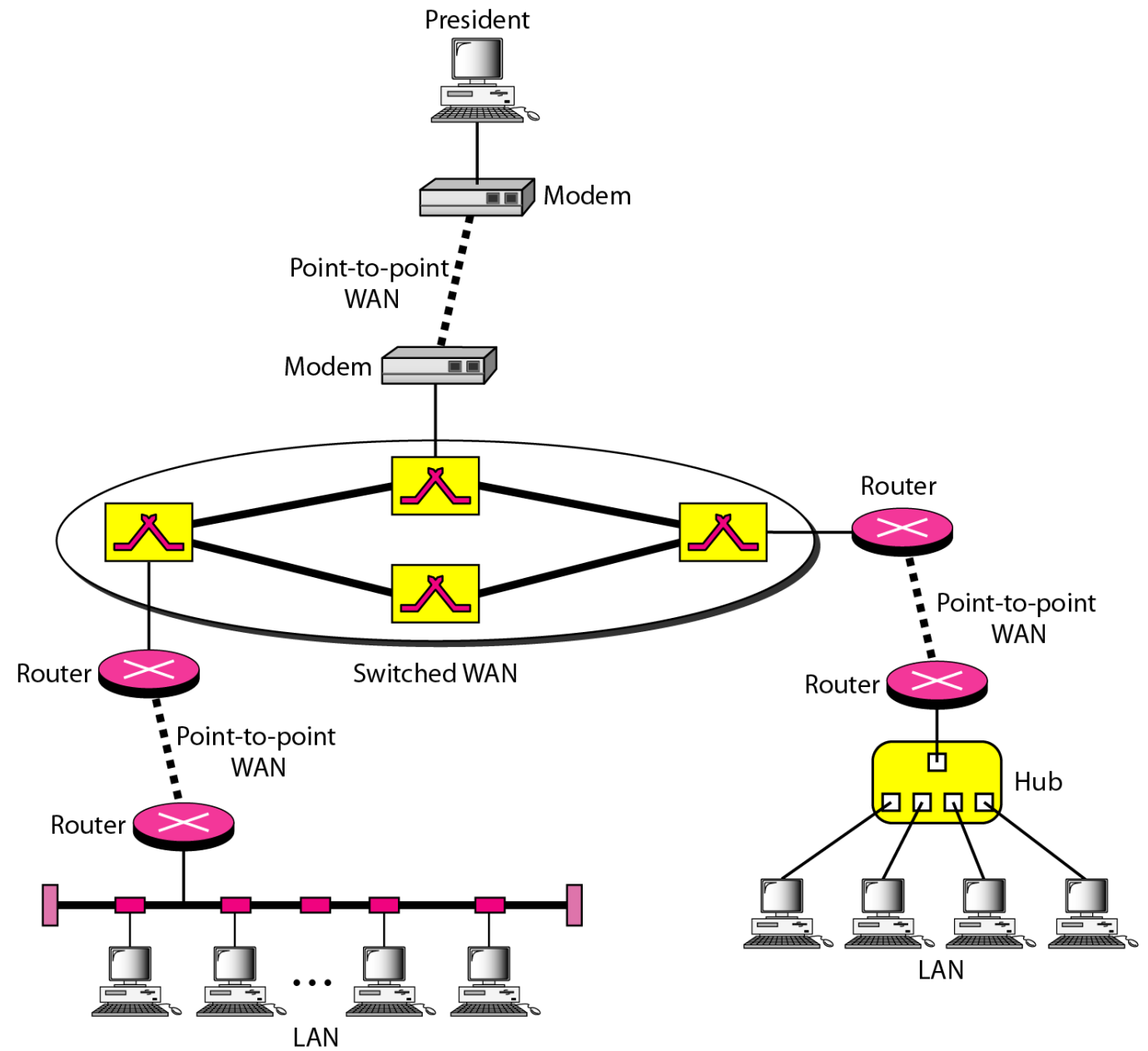
- Generally covers a city
- E.g. cable television network
- Came in existence around late 1990s to overcome poor over-the-air television reception
- Initially locally designed and ad hoc system and later on business companies entered in this field and set up wiring in entire city
- In Internet era this television network converted into two-way Internet service in unused part of the spectrum
- High speed wireless Internet access is also an example of MAN which has been standardized as IEEE 802.16

Wide Area Network

- A network that covers a **large geographical** area comprising a region, a country, a continent or even the whole world.
- WAN includes the technologies to transmit data over long distances and among different LANs and MANs.
- Communication links are provided by **public carriers** like telephone networks, network providers, cable systems, satellites etc.
- They generally have a **higher bit error rate**.
- Typically, they have **low data transfer rate** and **high propagation delay**, i.e., they have low communication speed.



Example:



Wide Area Network

- Spans a large geographical area often country or continent
- Single machine or LANs can be connected to other single machine or LANs through **communication subnet**
- Communication subnet typically owned and operated by a telephone company or Internet Service Provider
- Subnet consist of two components:
 - Transmission lines
 - Move bits between machines
 - Can be made of copper wires, optical fibers or even radio links
 - Switching elements
 - Specialized computers that connect three or more transmission lines
 - Determines outgoing lines for incoming data
 - Generally known as **router**

Wide Area Network

- Subnet consist of two components:
 - Transmission lines
 - Move bits between machines
 - Can be made of copper wires, optical fibers or even radio links
 - Switching elements
 - Specialized computers that connect three or more transmission lines
 - Determines outgoing lines for incoming data
 - Generally known as **router**
 - Router may store incoming packets and transmit them once output lines become free. Subnet using such mechanism is called a **store-and-forward** or **packet-switched** subnet.
 - Routing algorithm used to route packets
 - In WAN making use of satellite system, routers use antenna for transmission
 - Satellite networks are inherently broadcast

Wireless Networks

- Broadly there are three categories:
 - System Interconnection
 - Wireless LANs
 - Wireless WANs
- System Interconnection
 - Interconnecting network devices using short-range radio known as **Bluetooth**
 - E.g. connects computers, monitor, mouse, printer, digital cameras, headsets, scanners etc
 - No need of cables, driver installation
 - Big advantage: ease of use, just turn on and it works

Wireless Networks

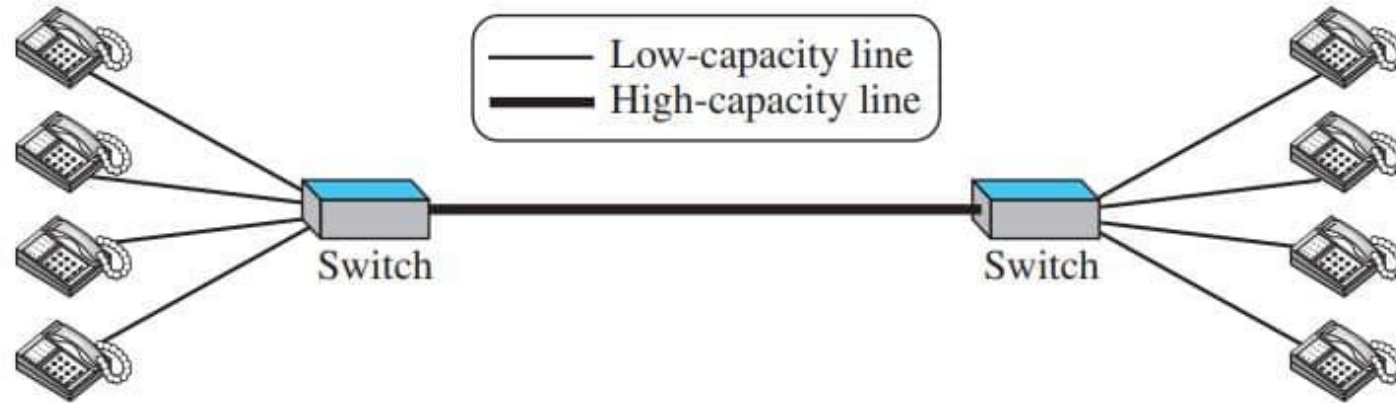
- Wireless LANs
 - Every computers has a radio modem and antenna with which it can communicate with other systems
 - Often there is an antenna on the ceiling that the machine talk to, however if the systems are close enough, they can communicate directly with one another in peer-to-peer configuration
 - Becoming increasingly popular in small offices and home networks
 - IEEE 802.11 is the standard for wireless LANs
- Wireless WANs
 - Radio network used for cellular telephone is an example of a low-bandwidth wireless network
 - High speed wireless network had initial focus on high-speed wireless Internet access from homes and offices bypassing the telephone system
 - IEEE 802.16 developed for the same

Switching

- Switch connects at least 2 links together
- Switch forwards data from one network segment to another
- Types
 - Circuit switched network
 - Packet switched network

Switching

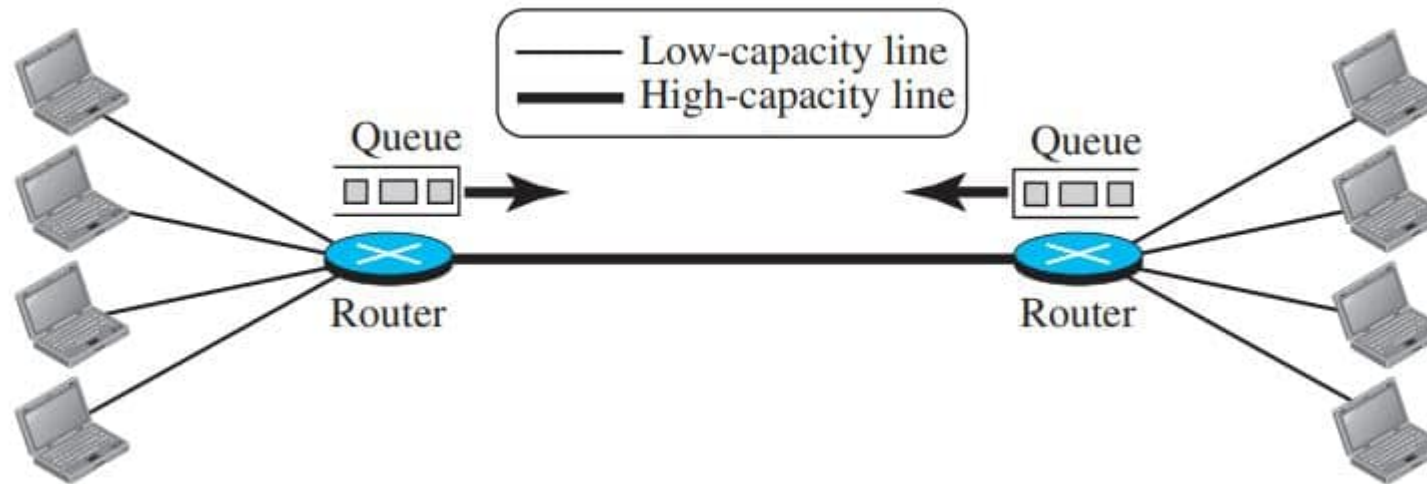
Circuit Switched network



Switch does not have storing capability; it only forwards

Switching

Packet Switched network



Router is used for both storing and forwarding as the packet is an independent entity

Internet

- Internet is a collection of networks or network of networks.
- Various networks such as LAN and WAN connected through suitable hardware and software to work in a seamless manner.
- The basic difference between WAN and Internet is that WAN is owned by a single organization while internet is not so.

Accessing Internet

- Using telephone networks
 - Dial-up Services
 - DSL Service
- Using Cable Network
- Using Wireless Networks
- Direct Connection to the Internet

Questions!!!